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■ Education

- BS Chemical Engineering, University of Illinois at Urbana-Champaign, 2009
- MS Predictive Analytics, Northwestern University, 2016

■ Career @ Dow

- **Product Development Intern**, Copolymers group, Dow Corning, summer 2008
- **Process R&D Engineer**, Copolymers group, Dow Corning, 2009 – 2012
- **High Throughput Research Engineer**, Dow Corning/Dow, 2012 – 2017
- **Data Scientist**, Manufacturing Chemometrics and AI, Dow, 2017 – 2021
- **Senior Data Scientist**, Information Research, Dow, 2021 – present
 - ✓ Research: Formulation modeling and inverse design
 - [The Digital Transformation of Product Formulation: Concepts, Challenge](#)
 - ✓ Model Deployment and MLOps Strategy Development
 - ✓ Citizen Data Science and Digital Transformation
 - ✓ GenAI and agentic research



Data Stewardship

Preserving the data you generate so it keeps paying off after your project ends

Alix Schmidt, Dow · Presenter

With content contributions from Peter Margl, Matt Benedict

Created using Claude Code

Key ideas for good data stewardship

1. Structure

What separates data a computer can read from data that needs an afternoon of cleaning.

Rectangular, tidy, typed

2. Identity

How a value in one row points at an entity outside the row, and how this forms the basis for advanced analysis.

IDs, join keys, canonical forms

3. Packaging

What turns a folder of files into a thing a stranger can use: a dataset with a dictionary, provenance metadata, and the code that produced it.

Data + code + docs = one asset

PART 1

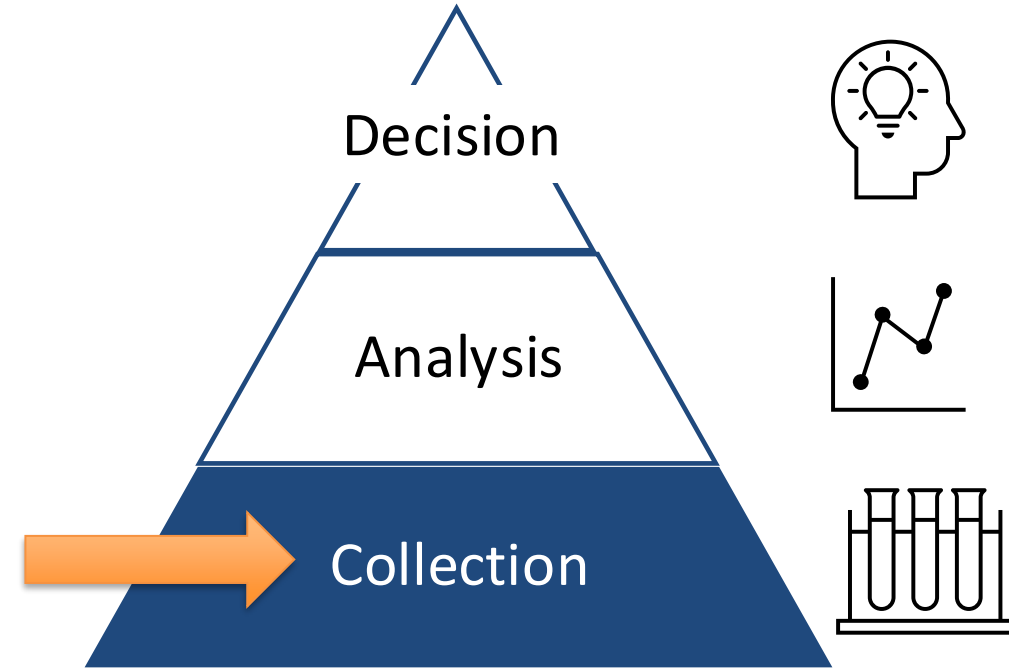
Why do we care about re-usable data?

An industry perspective on what makes a dataset valuable

DEFINITION

Data stewardship

A skill set for **creating, handling, and maintaining structured data** so it can be used for transparent, collaborative research decisions based on rigorous analyses.



FOR EXAMPLE

In practice: the discipline that turns 'the data I collected for my project' into 'the data the next three projects will build on.'

Why data stewardship matters

A PhD project

- **Starts with you**

you frame the question, you generate the data, you own the interpretation

- **Ends with you**

when the thesis ships, the dataset stops being touched, often within months

- **Half-life is short**

very few people ever revisit the raw numbers after the defense

- **Reproducibility is a courtesy**

not usually a requirement of anyone else's downstream work

An industry R&D project

- **Joins the pile**

every dataset compounds into institutional knowledge

- **Outlives the project**

the next project reads yours as its starting point, sometimes years later

- **Half-life measured in years**

long enough that people, tools, and file formats all turn over

- **Reproducibility is a job**

someone else will need to re-run your analysis and get the same number

TAKE A MOMENT

**Have you ever inherited someone's data
and been able to fully reproduce their
analysis in under one hour?**

Well-structured data supports four key principles of scientific research at once

Fidelity

Structured data supports data fidelity, which means it is **easy to avoid or detect errors** during the process of noting or storing data

Transparency

Structured data fosters collaboration because it is **easily understandable**

Reproducibility

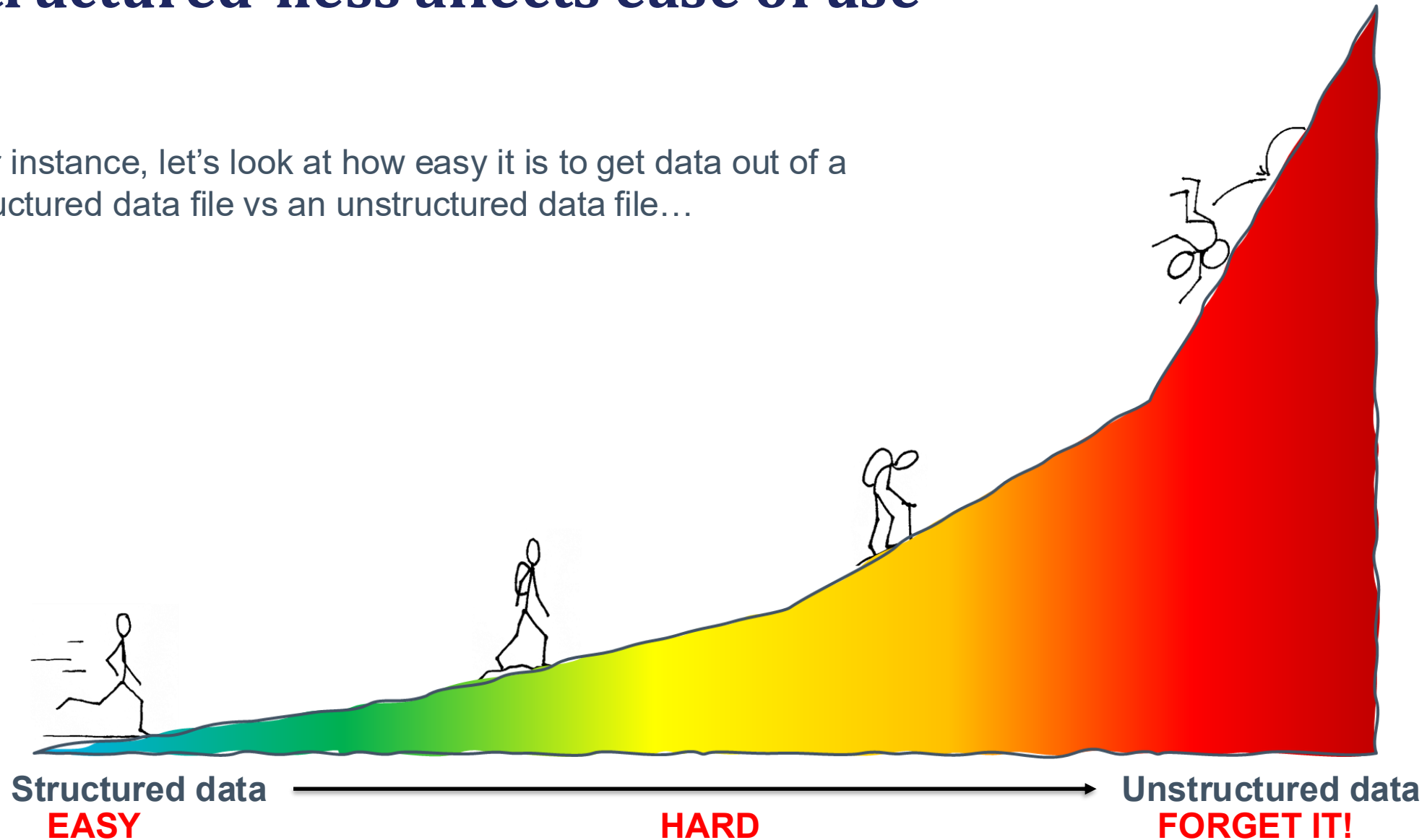
Structured data makes it easy for your colleagues to understand and **reproduce your results**

Collaboration

Transparency and reproducibility along with **good documentation** make collaboration on project data easy

Structured-ness affects ease of use

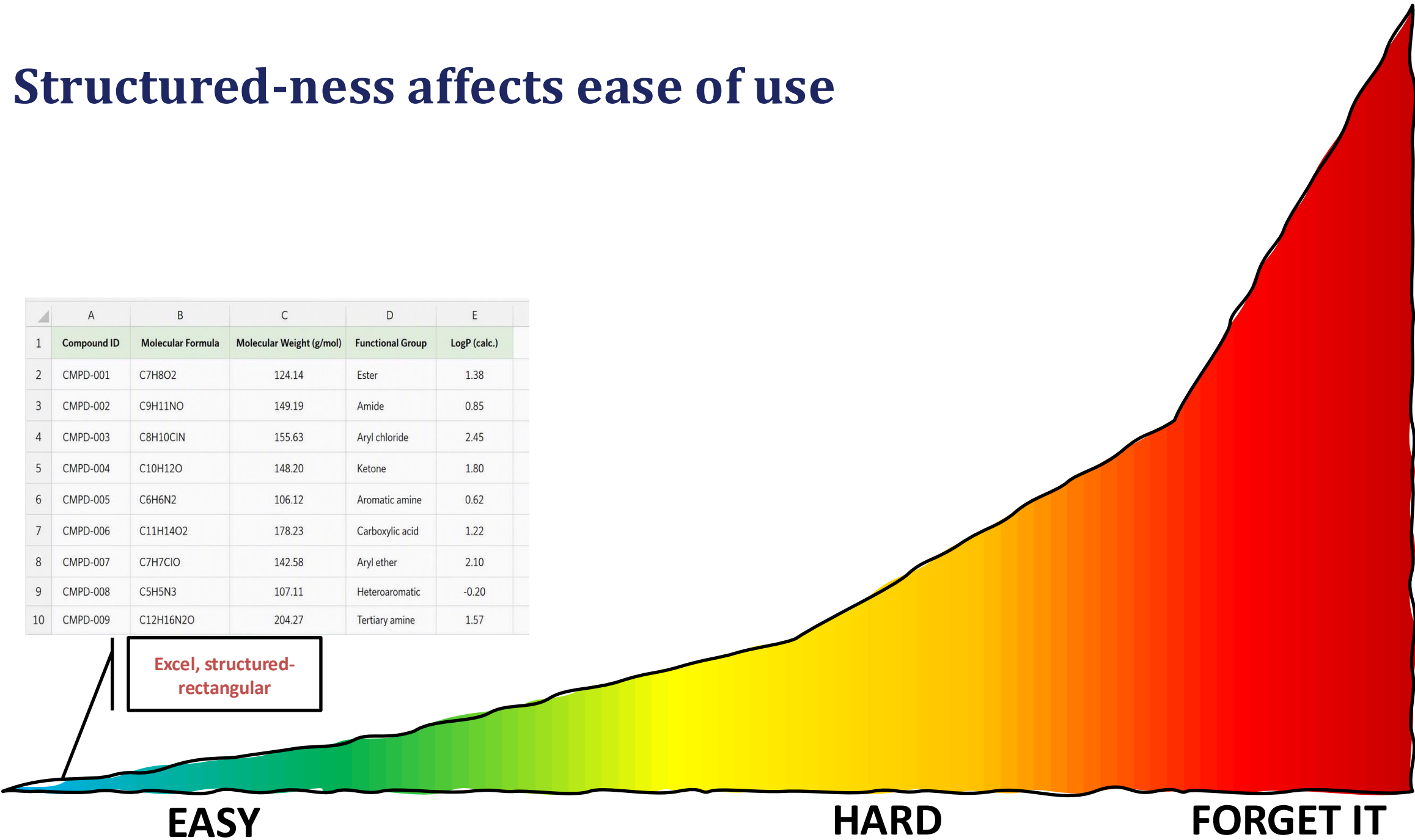
For instance, let's look at how easy it is to get data out of a structured data file vs an unstructured data file...



Structured-ness affects ease of use

	A	B	C	D	E	
1	Compound ID	Molecular Formula	Molecular Weight (g/mol)	Functional Group	LogP (calc.)	
2	CMPD-001	C7H8O2	124.14	Ester	1.38	
3	CMPD-002	C9H11NO	149.19	Amide	0.85	
4	CMPD-003	C8H10ClN	155.63	Aryl chloride	2.45	
5	CMPD-004	C10H12O	148.20	Ketone	1.80	
6	CMPD-005	C6H6N2	106.12	Aromatic amine	0.62	
7	CMPD-006	C11H14O2	178.23	Carboxylic acid	1.22	
8	CMPD-007	C7H7ClO	142.58	Aryl ether	2.10	
9	CMPD-008	C5H5N3	107.11	Heteroaromatic	-0.20	
10	CMPD-009	C12H16N2O	204.27	Tertiary amine	1.57	

Excel, structured-
rectangular



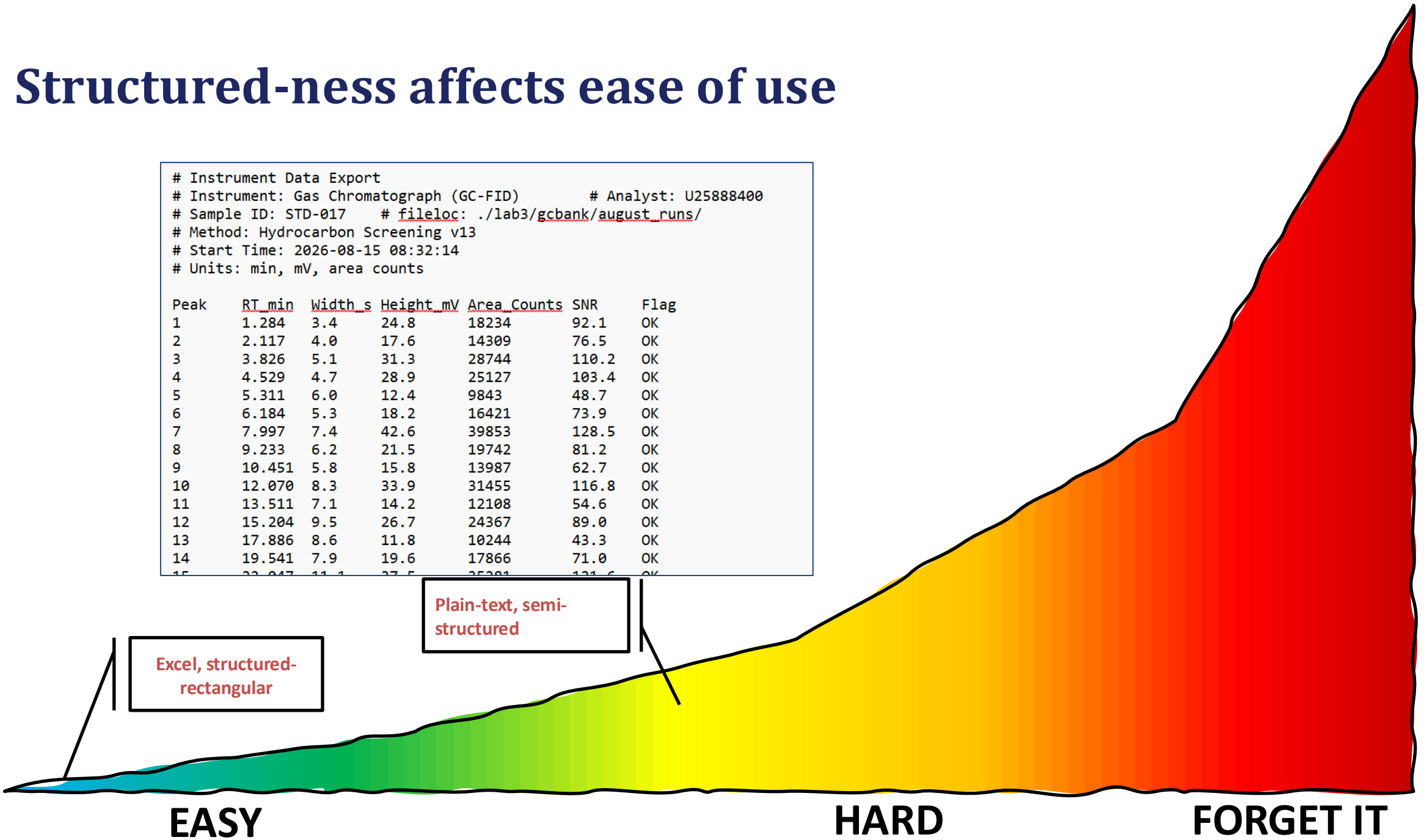
Structured-ness affects ease of use

```
# Instrument Data Export
# Instrument: Gas Chromatograph (GC-FID)      # Analyst: U25888400
# Sample ID: STD-017      # fileloc: ./lab3/gcbank/august_runs/
# Method: Hydrocarbon Screening v13
# Start Time: 2026-08-15 08:32:14
# Units: min, mV, area counts
```

Peak	RT_min	Width_s	Height_mV	Area_Counts	SNR	Flag
1	1.284	3.4	24.8	18234	92.1	OK
2	2.117	4.0	17.6	14309	76.5	OK
3	3.826	5.1	31.3	28744	110.2	OK
4	4.529	4.7	28.9	25127	103.4	OK
5	5.311	6.0	12.4	9843	48.7	OK
6	6.184	5.3	18.2	16421	73.9	OK
7	7.997	7.4	42.6	39853	128.5	OK
8	9.233	6.2	21.5	19742	81.2	OK
9	10.451	5.8	15.8	13987	62.7	OK
10	12.070	8.3	33.9	31455	116.8	OK
11	13.511	7.1	14.2	12108	54.6	OK
12	15.204	9.5	26.7	24367	89.0	OK
13	17.886	8.6	11.8	10244	43.3	OK
14	19.541	7.9	19.6	17866	71.0	OK
15	22.047	11.1	27.5	25281	121.6	OK

Plain-text, semi-structured

Excel, structured-rectangular



Structured-ness affects ease of use

Sigma-Aldrich. www.sigmaaldrich.com

SAFETY DATA SHEET Version 6.8
Revision Date 02/07/2023
Print Date 02/25/2023

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifiers
Product name : Benzene

Product Number : 319953
Brand : SIGALD
Index-No. : 601-020-00-8
CAS-No. : 71-43-2

1.2 Relevant identified uses of the substance or mixture and uses advised against
Identified uses : Laboratory chemicals, Synthesis of substances

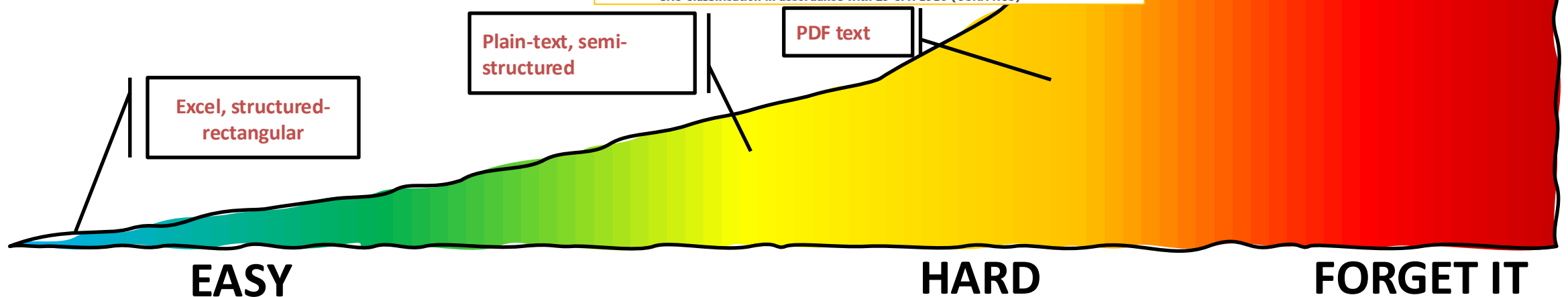
1.3 Details of the supplier of the safety data sheet
Company : Sigma-Aldrich Inc.
3050 SPRUCE ST
ST. LOUIS MO 63103
UNITED STATES

Telephone : +1 314 771-5765
Fax : +1 800 325-5052

1.4 Emergency telephone
Emergency Phone # : 800-424-9300 CHEMTREC (USA) +1-703-527-3887 CHEMTREC (International) 24 Hours/day; 7 Days/week

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture
GHS Classification in accordance with 29 CFR 1910 (OSHA HCS)



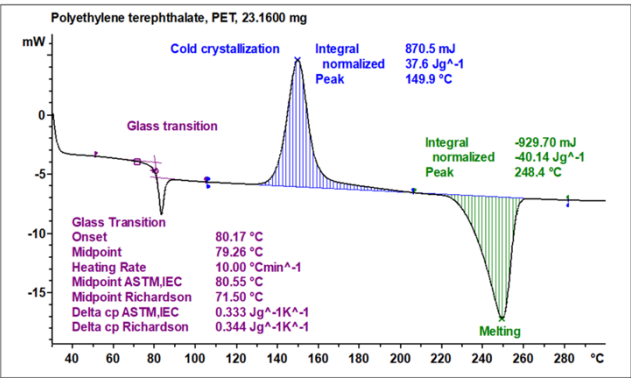
Structured-ness affects ease of use

mixed text & graphics,
PowerPoint or Excel

Latest results from pilot campaign

CDS

Very promising!!!



Dow

Plain-text, semi-structured

PDF text

Excel, structured-rectangular

EASY

HARD

FORGET IT

PART 2

Guidelines for Tidy Data

Rules for rectangular shapes, cells, types, identifiers, relationships

Rectangular · tidy · normalized

Three vocabularies for the same underlying discipline: one variable per column, one observation per row, one table per kind of entity. Rectangular is the shape. Tidy is the semantics. Normalized is the database term for splitting data across tables so no fact gets duplicated.

FOR EXAMPLE

Reactions in one table, catalyst reference in another, solvent reference in a third. You link them by shared identifiers, not by copying the catalyst's properties down every row of the reactions log.

Meet the Engine Oil Dataset

An employee at a company that makes formulated engine oils in multiple grades measures the viscosity of various oil grades coming off the production lines at various temperatures and records the data in a table. Each sample has a unique “sample number”.

sample_number	temperature:deg_c	viscosity:cp	material:sae_grade
SAM-AB1234	20.0	286	15W-40
SAM-AB1235	30.0	154	15W-40
SAM-AB1236	20.0	210	10W-40
SAM-AB1237	30.0	121	10W-40
SAM-AB1238	20.0	382	10W-60
SAM-AB1241	30.0	221	10W-60
SAM-AB1242	40.0	137	10W-60
SAM-AB1243	50.0	90	10W-60
SAM-AB1244	20.0	205	5W-40
SAM-AB1245	30.0	121	5W-40
SAM-AB1246	40.0	79	5W-40
SAM-AB1247	50.0	51	5W-40

Tidy data: rectangular shape

sample_number	temperature: deg_c	viscosity: cp	material: sae_grade
SAM-AB1234	20.0	286	15W-40
SAM-AB1235	30.0	154	15W-40
SAM-AB1236	20.0	210	10W-40
SAM-AB1237	30.0	121	10W-40
SAM-AB1238	20.0	382	10W-60
SAM-AB1241	30.0	221	10W-60

Rectangular data can be easily understood by humans as well as computers.

Rectangular data is a **prerequisite for processing and analysis of the data by computers**

Tidy data: Cells

✓ Do

- + one row per measurement**
no averages mixed in with individuals; no summary rows
- + empty cell for missing data**
blank means missing
- + annotations get their own column**
observation_notes column, not a comment or coloring key

✗ Don't

- merged cells**
each cell holds one value; a script reads cells, not layouts
- mixed data types**
'80 °C' is a string; '80' with a unit column is a number
- codes for missing values**
no 'n/a', 'N/A', '—', '999' — those are values, not blanks

sample_number	temperature: deg_c	viscosity: cp	material: sae_grade	observation_notes
SAM-AB1234	20.0		15W-40	~0.5 degC temperature drift
SAM-AB1235	30.0	154	15W-40	none
SAM-AB1236	20.0	210	10W-40	none
SAM-AB1237	30.0		10W-40	not enough sample
SAM-AB1238	20.0	382	10W-60	none

Tidy data: Identifiers

One column can uniquely name a row, or point to a thing outside the row.

Row identifier

unique per row — `sample_number`, `reaction_id`, `batch_id`. This is what makes a row addressable.

Entity identifier

points to a thing outside the table — `catalyst_id`, `operator_id`, `material_grade`. This is what makes tables joinable.

Canonical form, not free-typed

'PD_PPH3_4' spelled consistently, not 'Pd(PPh3)4' and 'pd(pph3)_4' and 'Pd(PPh₃)₄' all referring to the same catalyst.

Why are data Identifiers and data Relationships important?

Identifiers

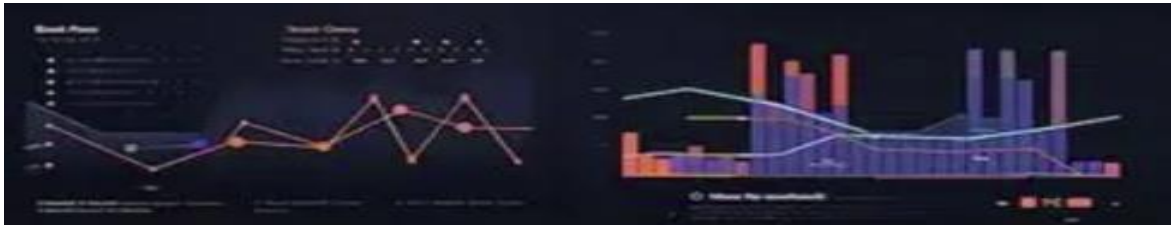
Identifiers are values in a data set that uniquely identify a particular thing

+

Relationships

Relationships stitch different pieces of data together to allow greater insights

Why you care about this...



	OrderDate	Region	Rep	Item	Units	Unit Cost	Total	OrderDate	Region	Rep	Item	Units	Unit Cost	Total	OrderDate	Region	Rep	Item	Units	Unit Cost	Total
1	1/6/15	East	Jones	Pencil	95	1.99	189.05	1/6/15	East	Jones	Pencil	95	1.99	189.05	1/6/15	East	Jones	Pencil	95	1.99	189.05
2	1/23/15	Central	Kivell	Binder	50	19.99	999.50	1/23/15	Central	Kivell	Binder	50	19.99	999.50	1/23/15	Central	Kivell	Binder	50	19.99	999.50
3	2/9/15	Central	Jardine	Pencil	36	4.99	179.64	2/9/15	Central	Jardine	Pencil	36	4.99	179.64	2/9/15	Central	Jardine	Pencil	36	4.99	179.64
4	2/26/15	Central	Gill	Pen	27	19.99	539.73	2/26/15	Central	Gill	Pen	27	19.99	539.73	2/26/15	Central	Gill	Pen	27	19.99	539.73
5	3/15/15	West	Sorvino	Pencil	56	2.99	167.44	3/15/15	West	Sorvino	Pencil	56	2.99	167.44	3/15/15	West	Sorvino	Pencil	56	2.99	167.44
6	4/1/15	East	Jones	Binder	60	4.99	299.40	4/1/15	East	Jones	Binder	60	4.99	299.40	4/1/15	East	Jones	Binder	60	4.99	299.40
7	4/18/15	Central	Andrews	Pencil	75	1.99	149.25	4/18/15	Central	Andrews	Pencil	75	1.99	149.25	4/18/15	Central	Andrews	Pencil	75	1.99	149.25
8	5/5/15	Central	Jardine	Pencil	90	4.99	449.10	5/5/15	Central	Jardine	Pencil	90	4.99	449.10	5/5/15	Central	Jardine	Pencil	90	4.99	449.10
9	5/22/15	West	Thompson	Pencil	32	1.99	63.68	5/22/15	West	Thompson	Pencil	32	1.99	63.68	5/22/15	West	Thompson	Pencil	32	1.99	63.68
10	6/8/15	East	Jones	Binder	60	8.99	539.40	6/8/15	East	Jones	Binder	60	8.99	539.40	6/8/15	East	Jones	Binder	60	8.99	539.40
11	6/25/15	Central	Morgan	Pencil	90	4.99	449.10	6/25/15	Central	Morgan	Pencil	90	4.99	449.10	6/25/15	Central	Morgan	Pencil	90	4.99	449.10
12	7/12/15	East	Howard	Binder	29	1.99	57.71	7/12/15	East	Howard	Binder	29	1.99	57.71	7/12/15	East	Howard	Binder	29	1.99	57.71
13	7/29/15	East	Parent	Binder	81	19.99	1,619.19	7/29/15	East	Parent	Binder	81	19.99	1,619.19	7/29/15	East	Parent	Binder	81	19.99	1,619.19
14	8/15/15	East	Jones	Pencil	35	4.99	174.65	8/15/15	East	Jones	Pencil	35	4.99	174.65	8/15/15	East	Jones	Pencil	35	4.99	174.65

OrderDate	Region	Rep	Item	Units	Unit Cost	Total	OrderDate
1/6/15	East	Jones	Pencil	95	1.99	189.05	1/6/15
1/23/15	Central	Kivell	Binder	50	19.99	999.50	1/23/15
2/9/15	Central	Jardine	Pencil	36	4.99	179.64	2/9/15
2/26/15	Central	Gill	Pen	27	19.99	539.73	2/26/15
3/15/15	West	Sorvino	Pencil	56	2.99	167.44	3/15/15
4/1/15	East	Jones	Binder	60	4.99	299.40	4/1/15
4/18/15	Central	Andrews	Pencil	75	1.99	149.25	4/18/15
5/5/15	Central	Jardine	Pencil	90	4.99	449.10	5/5/15
5/22/15	West	Thompson	Pencil	32	1.99	63.68	5/22/15
6/8/15	East	Jones	Binder	60	8.99	539.40	6/8/15
6/25/15	Central	Morgan	Pencil	90	4.99	449.10	6/25/15
7/12/15	East	Howard	Binder	29	1.99	57.71	7/12/15
7/29/15	East	Parent	Binder	81	19.99	1,619.19	7/29/15
8/15/15	East	Jones	Pencil	35	4.99	174.65	8/15/15

Rep	Item	Units	Unit Cost	Total
Jones	Pencil	95	1.99	189.05
Kivell	Binder	50	19.99	999.50
Jardine	Pencil	36	4.99	179.64
Gill	Pen	27	19.99	539.73
Sorvino	Pencil	56	2.99	167.44
Jones	Binder	60	4.99	299.40
Andrews	Pencil	75	1.99	149.25
Jardine	Pencil	90	4.99	449.10
Thompson	Pencil	32	1.99	63.68
Jones	Binder	60	8.99	539.40

Region	Rep	Item	Units	Unit Cost	Total	OrderDate
East	Jones	Pencil	95	1.99	189.05	1/6/15
Central	Kivell	Binder	50	19.99	999.50	1/23/15
Central	Jardine	Pencil	36	4.99	179.64	2/9/15
Central	Gill	Pen	27	19.99	539.73	2/26/15
West	Sorvino	Pencil	56	2.99	167.44	3/15/15
East	Jones	Binder	60	4.99	299.40	4/1/15
Central	Andrews	Pencil	75	1.99	149.25	4/18/15
Central	Jardine	Pencil	90	4.99	449.10	5/5/15
West	Thompson	Pencil	32	1.99	63.68	5/22/15
East	Jones	Binder	60	8.99	539.40	6/8/15

Data identifier examples

sample_number	temperature: deg_c	viscosity: cp	material: sae_grade
SAM-AB1234	20.0	286	15W-40
SAM-AB1235	30.0	154	15W-40
SAM-AB1236	20.0	210	10W-40
SAM-AB1237	30.0	121	10W-40
SAM-AB1238	20.0	382	10W-60
SAM-AB1241	30.0	221	10W-60

Identifiers are values in columns that uniquely identify a particular thing.

They can identify either a row in the table or a thing outside the table

Examples:

- a material
- a person
- another piece of information
- a row in a table

Knowledge check: Identifiers

Which of the columns in this table contain Identifiers?

A	B	C	D	E	F
sample_id	polymer_loading: wt%	solvent	temperature: degC	absorption: %	experimenter: user_id
402212	5.0	water	22.1	83	uc91123
402213	7.0	water	27.5	52	uc91123
406935	5.1	dmso	21.9	1	ub91244
406936	7.4	dmso	26.2	0	ub91244

What are data Relationships?

When an identifier points to another piece of information, a data relationship is created. The purpose of data relationships is to stitch different pieces together to allow greater insights. In the case below, the table on the right can be used to determine if a measurement in the left table was between the lower and upper limits for the grade in the left-hand side table because the material identifiers are shared.

sample_number	temperature: deg_c	viscosity: cp	material: sae_grade		material: sae_grade	temperature: deg_c	viscosity_upper_limit: cp	viscosity_lower_limit: cp
SAM-AB1234	20.0	286	15W-40		15W-40	20	290	285
SAM-AB1235	30.0	154	15W-40		10W-40	20	210	205
SAM-AB1236	20.0	210	10W-40		10W-60	20	377	382
SAM-AB1237	30.0	121	10W-40		...	...	...	...
SAM-AB1238	20.0	382	10W-60					
SAM-AB1241	30.0	221	10W-60					
...	...	...	...					

Fictitious data

Relationships work if identifiers are shared

sample_number	temperature: deg_c	viscosity: cp	material: sae_grade		material: sae_grade	temperature: deg_c	viscosity_upper_limit: cp	viscosity_lower_limit: cp
SAM-AB1234	20.0	286	15W-40	→	15W-40	20	290	285
SAM-AB1235	30.0	154	15W-40		10W-40	20	210	205
SAM-AB1236	20.0	210	10W-40	→	10W-40	20	377	382
SAM-AB1237	30.0	121	10W-40		...	...	...	...
SAM-AB1238	20.0	382	10W-60	→	10W-60			
SAM-AB1241	30.0	221	10W-60					
...	...	...	...					

Fictitious data

Column names don't need to match!
(but it's very helpful if they do)

Relationships work if identifiers are shared

But the identifiers themselves must match

sample_number	temperature: deg_c	viscosity: cp	material: sae_grade		material: sae_grade	temperature: deg_c	viscosity_upper_limit: cp	viscosity_lower_limit: cp
SAM-AB1234	20.0	286	15W40	✗	15W-40	20	290	285
SAM-AB1235	30.0	154	15W40		10W-40	20	210	205
SAM-AB1236	20.0	210	10W-40		10W-60	20	377	382
SAM-AB1237	30.0	121	10W-40		...	...	...	...
SAM-AB1238	20.0	382	10W-60					
SAM-AB1241	30.0	221	10W-60					
...	...	...	...					

Fictitious data

Even small spelling differences will break the join

Knowledge check: Data relationships

Which of the following pairs of tables enables a data relationship?

A

sample_id	sample_weight: g	material: name
PMM373	312.1	toluene
JDA783	531.5	hexane

material: cas_number	molecular_weight: g/mol
108-88-3	92.14
110-54-3	86.18

B

sample_id	sample_weight: g	material: casno
PMM373	312.1	108-88-3
JDA783	531.5	110-54-3

material: cas_number	molecular_weight: g/mol
108-88-3	92.14
110-54-3	86.18

C

sample_id	sample_weight: g	material: name
PMM373	312.1	Peter_sample_1
JDA783	531.5	hexane

material: name	molecular_weight: g/mol
Peter_sample1	92.14
n-hexane	86.18

PART 3

Packaging: from files to data products

What turns a folder of files into something a stranger can pick up and use

A re-usable digital asset has three legs

Data

The numbers and observations, arranged so someone else can load them into their tool of choice without hand-transcription. This is what most people mean when they say 'data', but it's only one piece of complete data stewardship.

csv · tidy xlsx · sql

Code

The transformations, models, plots, and joins that produced the outputs. A report of numbers with no script that made them is irreproducible; nobody can re-derive anything.

code · git · requirements.txt

Docs

What, *exactly*, the columns mean, what units apply, who collected the data, what instrument was used, what method was followed.

dictionary · README · provenance

A data dictionary — one row per column of your dataset

column_name	type	unit	allowed_values	description
reaction_id	str	—	R001-R999	<i>unique per row; assigned at logging time</i>
catalyst_id	str	—	PD_PPH3_4 PD_C	<i>canonical catalyst identifier (see cat table)</i>
temperature_c	float	°C	20 - 200	<i>setpoint on reactor jacket at start of run</i>
yield_pct	float	%	0 - 100	<i>isolated yield after column chromatography</i>
run_date	date	—	YYYY-MM-DD	<i>day the run started</i>
operator_id	str	—	Dow user id	<i>who ran the experiment</i>

Minimum-viable fields: column name, type, unit, allowed values, description

Easy steps to self-documenting data

Column headers are explicit and clearly define what was recorded

“Temperature setpoint”, not “Tsp”

Column headers contain units clearly separated from name

“Temperature setpoint : degC”, not “ToC”

Use an agreed-upon material ID instead of product names

“ELN10234” instead of “Dave’s polymer 3”

One row per measurement, keep all measurements

no averaged data or hand-picked data points

Use separate columns for intended vs actual and name them accordingly

reactor set-point temperature vs actual measured reactor temperature

Refer to measurement procedures, if possible by including them in the column name

“ASTM_d2983-09_viscosity”

Dataset-level metadata — the questions the columns don't answer

PROVIDING METADATA GIVES FUTURE RESEARCHERS THE CONTEXT NEEDED TO RE-USE YOUR DATA

☐

Who collected it

user id or name — 'A. Schmidt, Materials Chemistry.' Not 'the lab.'

☐

When it was collected

specific dates — '2026-03-14 through 2026-03-21.' Not 'last quarter.'

☐

What instrument produced it

make, model, unit — 'Anton Paar MCR 302, s/n M172.' Calibration date if it matters.

☐

What settings applied

'10 mL/min flow · 80 °C · 60 min residence.' Reactor set-point and measured, if different.

☐

What method was used

reference the procedure by name — 'SOP-CAT-023 v2.1' — not by verbal description.

FAIR — a data interoperability framework

WILKINSON ET AL., SCIENTIFIC DATA 3, 160018 (2016)

[The FAIR Guiding Principles for scientific data management and stewardship | Scientific Data](#)

Findable

someone else can locate your dataset — persistent identifier, indexed in a catalog, not buried in your Google Drive.

Accessible

once found, the data can actually be retrieved — with clear terms about who's allowed and how. 'Email me for a copy' is not accessible.

Interoperable

the format and vocabulary can be read alongside other datasets — standard file formats, controlled vocabularies, machine-readable schemas.

Reusable

there's enough context — license, provenance, dictionary — for someone to use it responsibly in a new setting.

How to share data — and how not to

✓ Good ways

+ Shared file location

SharePoint, project file share, cloud bucket — everyone sees the same file

+ Project database

ELN, LIMS — schema enforced by the system, not by the user

+ Version-controlled repo

git for the code and data-generation scripts; changes are auditable

✗ Ineffective ways

— Non-rectangular Excel

tables scattered around a sheet, multi-header rows, formatting-as-data

— Tables inside PowerPoint

numbers stuck to a slide — a screenshot of data, not data

— Email, chat, PDFs

worst extraction task; each one becomes an archaeology project

Code deserves the same discipline as data

A bare script is like an unlabeled spreadsheet

A .py with no README, no requirements pin or venv, no example inputs. It ran on your machine last Tuesday; it does not run anywhere else.

A packaged script

code + requirements.txt (or pyproject.toml) + a short README describing inputs, outputs, and how to run it. Same three legs as a dataset: content + spec + docs.

Version control is a lab notebook for code

git tracks what changed, when, and why. If your only backup is 'final_v3.py' next to 'final_v3_ACTUALLY_final.py' — you don't have version control, you have a warning sign.

What to remember

Structure beats free-form

Rectangular, tidy, neat cells.

Identifiers are load-bearing

Identifiers build data relationships to enrich your analysis. Choose meaningful identifiers with your team and type them carefully.

Document the dataset

Use a data dictionary and dataset metadata to remove ambiguity about what your data means for future users.

Data + code + docs = one asset

Clearly documented data and the code used for transformations and processing leave a clear audit trail of your analysis.

TAKE A MOMENT

**What would you change about your
current data workflow starting next
week?**

References and further reading

- **Wickham, H.**

["Tidy Data." Journal of Statistical Software 59\(10\), 2014](#)

- **Broman & Woo**

["Data Organization in Spreadsheets." The American Statistician 72\(1\), 2018](#)

- **Wilkinson et al.**

["The FAIR Guiding Principles for scientific data management and stewardship." Scientific Data 3, 160018 \(2016\)](#)

- **Data Carpentry**

datacarpentry.org — open lessons, especially Data Organization in Spreadsheets